

**BIRLA INSTITUTE OF TECHNOLOGY,
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**Department of Electronics & Communication
Engineering**

PEST DETECTION AND IDENTIFICATION

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OUTLINE

- Introduction
- Objectives
- Motivation
- Literature Review
- Methodology
- Yolo v5.
- Results so far.
- Work done so far & Further work to be done
- References

Introduction

- Rice is the most important and a primary source of food in Asia. However, rice may lose its quantity and quality when rice is attacked by different insect pests. Therefore, it is a top priority to find effective methods to reduce the level of their infestation in the paddy fields.
- In agriculture, pest control has always been considered as the most challenging task for farmers. Most of the farmers used the traditional pest management methods which is the regular spray program based on schedules rather than the presence of insect pests on the paddy fields. These chemicals kill useful insects which eradicate pests in crops.
- Prior detection of pests in the image is important for control of pests in the fields. Due to this, classifying the pest in the images has been an onerous task.

- In traditional agricultural production, the way to obtain pest population dynamics in the field mainly relies on manual surveys. However, this approach is labor-intensive and has some obvious drawbacks: inefficiency, subjectivity, error-prone, and lagging information. For the sake of food security and yield, it is desirable to develop an automatic pest monitoring method with high efficiency and accuracy.
- Image processing is the analysis and manipulation of graphical images from sources such as photographs and videos. There are three main steps in image processing; first, is the conversion of captured images into binary values that a computer can process; second, is the image enhancement and data compression; and the third is the output step that consists of the display or printing of the processed image.

Objectives

These are some objectives of our project:

- (a) Pest detection and identification using YOLO v5.
- (b) Preventive and Control measures to avoid further crop damage.

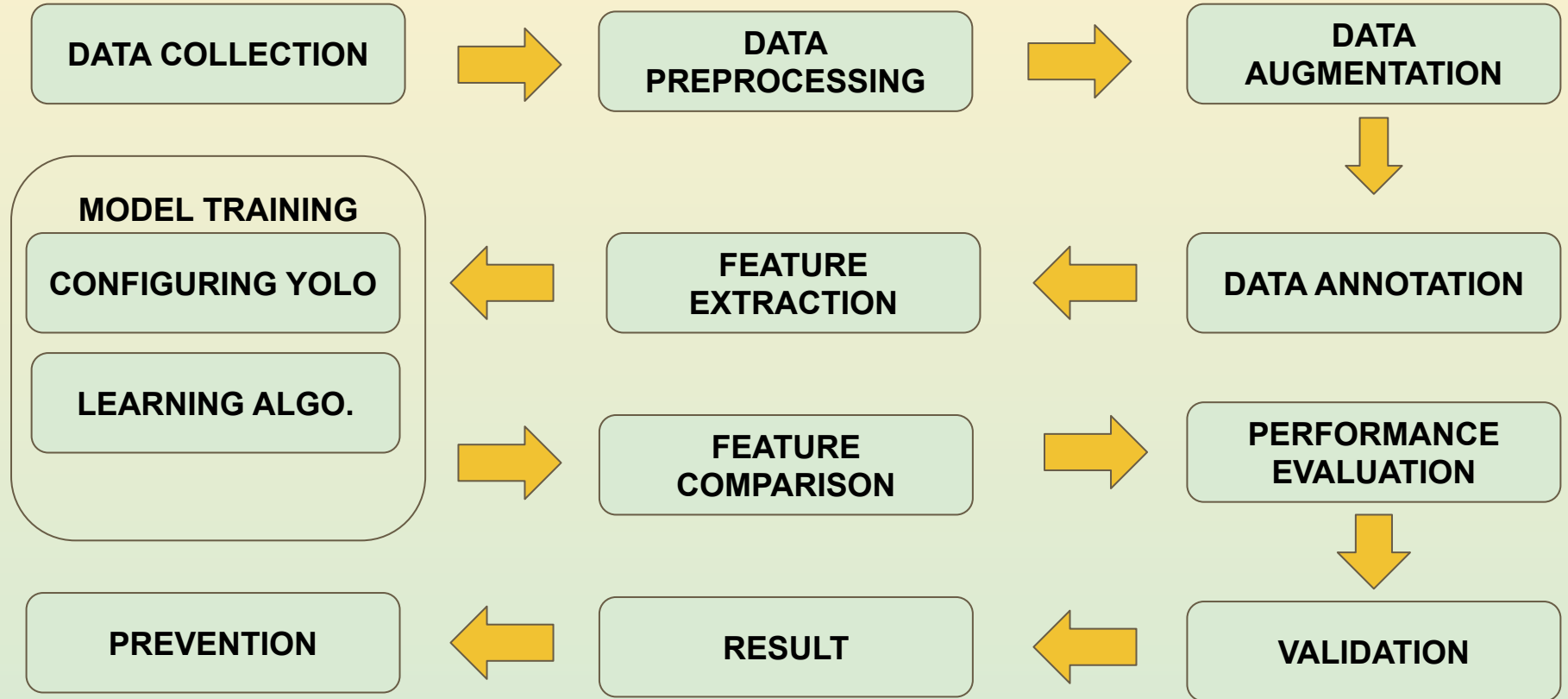
Motivation

- The idea of building this project came to our mind when we studied about the biggest threat to successful crop production i.e insect pests and diseases. These pests cause enormous amount of damage to crops which incurs huge losses on farmers economically.
- But early detection of these pests could reduce this damage upto 40%. Monitoring the crops manually is not possible.
- This is where our project comes in hand. It provides solutions to monitor crops faster and detect pests more accurately using machine learning model and deal with it as soon as possible in order to minimize the damage suffered by the crops.

Literature Review

AUTHOR	JOURNAL	ARTICLE	OBJECTIVE
Eray Önlér [1]	International Journal of Agricultural and Natural Sciences(2021)	Real Time Pest Detection Using YOLOv5	An object detection system to detect the thistle caterpillar (Vanessa cardui), that can cause damage to sunflower cultivation.
Hainie Zha,Yuan Liu ,Kui Liu , Jianfeng Liao, Jing Zhang, Pei Zhad , Jialong Zhou , Zhang ying Ye [2]	Social Science Research Network(2023)	Research on Rice Pest Classification and Identification	It proposes a classification and recognition model for small pest targets based on the combination of YOLO-v5
Suxuan Li , Zelin Feng , Baojun Yang , Hang Li , Fubing Liao , Yufan Gao , Shuhua Liu , Jian Tang, Qing Yao. [3]	Frontiers in Plant Science(2022)	An intelligent monitoring system of diseases and pests on rice canopy	Developed an intelligent monitoring system for detecting and identifying the disease and pest lesions on the rice canopy.
Jun Liu and Xuewei Wang.[4]	Frontiers in Plant Science(2020)	Tomato Diseases and Pests Detection Based on Improved Yolo V3 Convolutional Neural Network	To detect tomato pests with the help of deep learning and YOLOv3.

METHODOLOGY



- **Data Collection** : Collected through IP102-Dataset - Kaggle.Two classes - Yellow Rice Borer (453 images) and Rice Gall Midge (456 images). Total (909 images)
- **Data Preprocessing** : Data Resize, Auto Orient, Grey-Scale. (Roboflow)
- **Data Augmentation** : Increasing Volume to 2090 images by adjusting brightness and rotations. (Roboflow). (80% Training, 10% Testing, 10% Validating)
- **Data Annotation** : Manually annotate and segmenting through V7 Darwin Tool.
- **Feature Extraction** : Transforming raw data into numerical features that can be processed while preserving the information in the original data set.
- **Model Training** : Configuring Yolo and training Algorithm in Python/Collab.
- **Feature Comparison** : Comparing features for test images for classification.
- **Performance Evaluation** : Evaluating the performance of model.
- **Validation** : Validating model through another set of images.
- **Result** : Concluding the project.
- **Preventive and Controlive Measures** : Suggesting measures for prevention.

YOLO v5

- YOLO is a state of the art, real-time object detection algorithm created by Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi in 2015 and was pre-trained on the COCO dataset. It uses a single neural network to process an entire image. The image is divided into regions and the algorithm predicts probabilities and bounding boxes for each region.
- YOLO is well-known for its speed and accuracy and it has been used in many applications like: healthcare, security surveillance and self-driving cars.
- The main difference between the architecture of YOLOv3, YOLOv4, and YOLO v5 is the Darknet53 backbone used by YOLOv3. The backbone of YOLOv4 architecture is CSPdarknet53, and YOLOv5 uses the focus structure of CSPDarknet53 as the backbone. In YOLO v5, the focus layer is added for the first time. The focusing layer swaps the original three layers of the YOLOv3 algorithm. The benefits of the focus layer include decreased memory requirements, reduction in layer weights, and onward and backward propagation optimization.

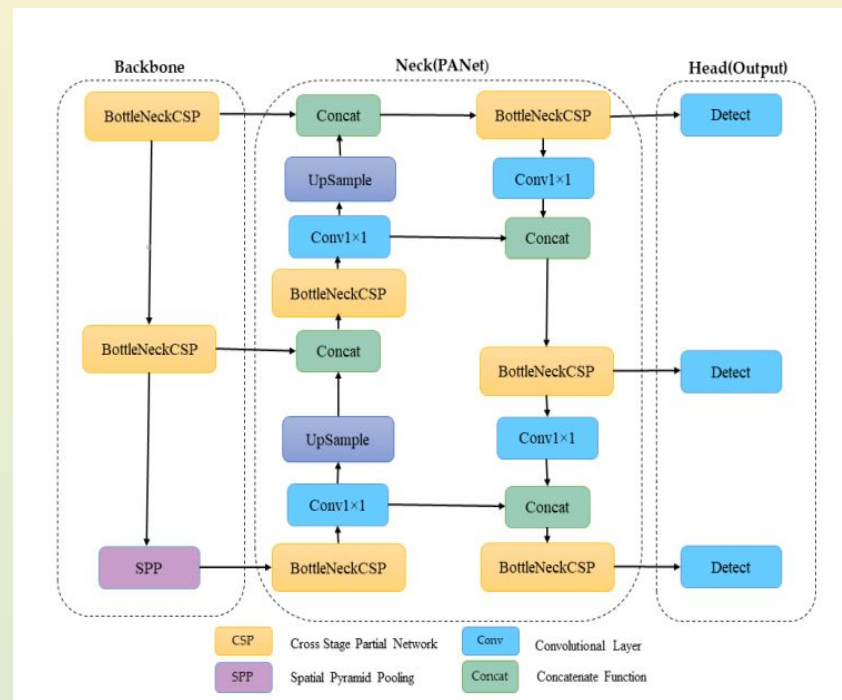
YOLO v5 Architecture

Yolo v5 consists of three parts:

- 1) Backbone: CSPDarknet
- 2) Neck: PANet
- 3) Head: Yolo Layer

The data are first input to CSP(Cross Stage Partial Network) Darknet for feature extraction, then fed to PANet for feature fusion.

Finally Yolo Layer outputs detection results (class,score, location, size).



Results so far (Dataset).

Yellow Rice Borer (Input images)



Rice Gall Midge (Input Images)



Instance Segmentation



成虫

Labels

YELLOW RICE BORER 1

Model output

```
{
  "bounding_box": {
    "h": 583,
    "w": 294,
    "x": 267,
    "y": 54
  }
}
```

Use another image



Labels

RICE GALL MIDGE 1

Model output

```
{
  "bounding_box": {
    "h": 437,
    "w": 623,
    "x": 96,
    "y": 94
  }
}
```

Use another image



RICE GALL MIDGE



RICE GALL MIDGE

雌成虫



RICE GALL MIDGE

雄成虫

食蚜瘿蚊

Anhidodeltes abietis (Kieffer)



YELLOW RICE BORER

♀

YELLOW RICE BORER

中国农视网 agnjw.net



YELLOW RICE BORER

YELLOW RICE BORER (Result Using Yolo v5)



RICE GALL MIDGE (Result Using Yolo v5).



Performance Evaluation

mAP - Mean Average Precision(mAP) is a metric used to evaluate object detection models such as Fast R-CNN, YOLO, Mask R-CNN, etc.mAP formula is based on the following sub metrics: Confusion Matrix, Intersection over Union(IoU), Recall, Precision.

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i$$

Precision - It is a metric that quantifies the number of correct positive predictions made. The formula to calculate precision is

$$Precision = \frac{TP}{TP + FP}$$

Where TP=True Positives and FP=False Positives.

Recall - It is a metric that quantifies the number of correct positive predictions made out of all the positive predictions that could have been made.

$$Recall = \frac{TruePositive}{TruePositive + FalseNegative}$$

The values obtained are as follows:-

mAP	Precision	Recall
91.9%	96.7%	77.3%

Work done so far

- Dataset collection
- Data Annotation
- Segmentation
- Understanding the theory related to yolo v5 and image processing.
- Training the model.

Further work to be done

- Validating the model by increasing the no. of classes.
- Validating the model by collecting Real Time Images of Pests of different classes.
- Performance Evaluation.
- Deployment
- Supportive Preventive and Control measures.

References

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THANK YOU